

## Multiple Choice

1. **Answer:** A  
*Options:* A) 0.5; B) 1; C) 0.95; D) 0.65; E) 0.85  
*Note:* Correct option: A - 0.5
2. **Answer:** E  
*Options:* A) 0.8; B) 0.3; C) 0.1; D) 0.5; E) 0.4  
*Note:* Correct option: E - 0.4
3. **Answer:** A  
*Options:* A) False, False; B) True, True; C) Not Determinable, True; D) Not Determinable, False; E) True, False  
*Note:* Correct option: A - False, False
4. **Answer:** C  
*Options:* A) \$4.52; B) \$7.28; C) \$5.52; D) \$6.28; E) \$3.72  
*Note:* Correct option: C - \$5.52
5. **Answer:** C  
*Options:* A) -1; B) -13; C) -4; D) -3; E) 0  
*Note:* Correct option: C - -4

## True / False

6. **Answer:** False  
*Options:* A) True; B) False  
*Note:* LLM-generated false statement. Correct answer: '0.5'.
7. **Answer:** True  
*Options:* A) True; B) False  
*Note:* LLM-generated true statement based on MCQ.
8. **Answer:** False  
*Options:* A) True; B) False  
*Note:* LLM-generated false statement. Correct answer: 'Statement 1 is true, Statement 2 is not applicable'.
9. **Answer:** False  
*Options:* A) True; B) False  
*Note:* LLM-generated false statement. Correct answer: '200'.

10. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: '126.26'.

## Long Form Response

11. **Answer:** The length of the hypotenuse is given by the distance formula to be  $\sqrt{(5 - (-1))^2 + (-5 - (-1))^2} = \sqrt{6^2 + 4^2} = \sqrt{52}$ . The length of the leg is then given by  $\sqrt{52}/\sqrt{2} = \sqrt{26}$  (alternatively, the Pythagorean Theorem can be applied), and the area of the isosceles right triangle is then equal to  $\frac{1}{2} \cdot \sqrt{26} \cdot \sqrt{26} = 13$ .

*Note:* Students should show all steps in their mathematical solution.

12. **Answer:** 100. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

*Note:* Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

*End of Answer Key*